

Commands Used in the P4 Mini Project

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1 Project Directory

```
cd /mnt/e/ATN/p4-mini-project
```

2 Environment Setup

2.1 Update System

```
sudo apt update && sudo apt upgrade -y
```

2.2 Install Dependencies

```
sudo apt install -y git build-essential cmake \  
python3 python3-pip python3-venv \  
libboost-all-dev libpcap-dev libssl-dev \  
bison flex pkg-config
```

2.3 Python Environment

```
python3 -m venv ~/p4env  
source ~/p4env/bin/activate
```

2.4 Install Mininet

```
sudo apt install -y mininet  
sudo mn --test pingall --switch user
```

2.5 Install BMv2

```
git clone https://github.com/p4lang/behavioral-model.git
cd behavioral-model
mkdir build && cd build
cmake ..
make -j1
sudo make install
```

2.6 Install p4c

```
git clone https://github.com/p4lang/p4c.git
cd p4c
mkdir build && cd build
cmake ..
make -j1
sudo make install
```

3 Compile P4 Program

```
sudo mn -c
mkdir -p build
p4c --target bmv2 --arch vlmodel basic_ml.p4 -o build
```

4 Run Project

```
bash run.sh test
bash run.sh
```

5 Run Topology

```
sudo /usr/bin/python3 topology.py
sudo /usr/bin/python3 topology.py --test
```

6 Controller Commands

```
python3 controller.py --dry-run
python3 controller.py
```

7 Mininet Commands

```
pingall
h1 ping -c 3 10.0.2.10
h2 tcpdump -enn -i eth0
h1 traceroute -n 10.0.2.10
```

8 Throughput Test

```
h2 iperf3 -s -1
h1 iperf3 -c 10.0.2.10 -t 3
```

9 Debugging

```
echo "table_dump ipv4_lpm" | simple_switch_CLI --thrift-port 9090
sudo ss -ltnp | grep 9090
sudo cat /tmp/p4s.s1.log
sudo tail -f /tmp/p4s.s1.log
```

10 Save Output

```
bash run.sh test 2>&1 | tee run_output.txt
```

11 Exit

```
exit
sudo mn -c
```

12 Expected Output

```
P4 switch s1 has been started.  
Entry has been added with handle 0  
Entry has been added with handle 1  
3 packets transmitted , 3 received , 0% packet loss  
64 bytes from 10.0.2.10: ttl=63  
iperf Done.
```